

SkillVerse

A Complete Learning Curriculum

Bash · Git · Makefile · OBI · RTL Design · RTL Verification
SystemVerilog · Unix Terminal · VIM · Vivado

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bash

1. Welcome to the World of Bash Scripting Mastery!

- Why Bash is Your Indispensable Skill
- Setting Up Your Bash Environment
- Your First Bash Script

2. Bash Scripting Essentials

- Your First Script
- Core Shell Commands
- Variables
- Operators
- Exercise
- Pro Tips for Bash Beginners (and Beyond!)

3. Control Structures

- Conditional Statements
- Loops
- case Statements
- Exercise
- Pro Tips for Mastering Control Structures

4. Functions in Bash

- Defining Functions
- Function Parameters
- Returning Values from Functions
- Exercise
- Pro Tips for Working with Bash Functions

5. Working with Files in Bash

- File Test Operators
- Reading and Writing Files
- File Permissions
- Directory Management
- File Handling Best Practices
- Enhanced Exercise

6. Advanced Scripting Techniques

- Regular Expressions (RegEx)
- Sed
- Awk
- Process Substitution
- Command Substitution
- Exercise

7. Error Handling in Bash Scripting

- Exit Status
- Trap Command
- Debugging Techniques
- Exercise

8. Practical Examples of Bash Scripting

- Backup Scripts
- Automation Scripts
- Parsing Logs
- System Monitoring Scripts
- Exercise

9. Best Practices for Shell Scripting

- Writing Readable Code
- Commenting and Documentation
- Script Optimization
- Error Handling
- Input Validation
- Use Functions for Reusability
- Always Quote Variables
- Include `set -euo pipefail`
- Exercise

git

1. Introduction to Git

- Understanding Git
- Why Git is the Industry Standard
- Installing Git on Your System
- Next Steps After Installation

2. Getting Started with Git

- Creating a New Git Repository via Command Line Interface (CLI)
- Creating a New Repository via GitHub Website
- Cloning a Git Repository

3. Core Git Concepts Explained

- Branch
- Commit
- Issue
- Push
- Pull
- Pull Request (PR)

4. Essential Git Commands

- `git init`
- `git clone`
- `git status`
- `git add`
- `git commit`
- `git push`
- `git fetch`
- `git pull`
- `git checkout`
- `git log`
- `git diff`
- `git reset`
- `git stash`

5. Mastering Git Merge Conflicts

- Understanding the Root Causes of Merge Conflicts
- Identifying Merge Conflicts
- Step-by-Step Guide to Resolving Merge Conflicts

- Pro Conflict Resolution Toolkit
- Real-World Scenario
- Merge Conflict Resolution Cheat Sheet
- Wisdom Nuggets for Conflict Wrangling

6. Git Workflow Guide

- Basic Git Workflow (Team Collaboration)
- Advanced Workflow (Open-Source Contribution)
- Workflow Comparison
- Best Practices for Both Workflows
- Common Pitfalls & Solutions
- Workflow Cheat Sheet

7. Level Up Your Git Skills

- Craft Commit Messages That Shine
- Branch Like a Git Guru
- Stay Synced Upstream
- Review Code Before It Lands
- Keep Your Repo Tidy
- Go the Extra Mile
- Bonus Git Power-Ups

Makefile

1. Introduction to Makefile

- What is Makefile?
- History and Importance of Makefile
- Basic Syntax and Structure
- Simple Makefile Example
- Simple Exercise

2. Variables and Functions

- Defining and Using Variables
- Built-in Functions
- String Manipulation Functions
- Conditional Functions
- Simple Exercise

3. Targets and Dependencies

- Understanding Targets
- Specifying Dependencies
- Writing Rules
- Phony Targets
- Simple Exercise

4. Pattern Rules and Wildcards

- Introduction to Pattern Rules
- Using Wildcards
- Automatic Variables
- Static Pattern Rules
- Simple Exercise

5. Advanced Makefile Features

- Include Directive
- Makefile Functions
- Conditional Directives
- Overriding and Appending Variables
- Simple Exercise

6. Debugging and Optimization

- Debugging Makefiles
- Common Errors and Troubleshooting

- Optimizing Makefiles for Performance
- Best Practices
- Simple Exercise

7. Real-world Examples and Projects

- Writing Makefiles for C/C++ Projects
- Using Makefile with Other Programming Languages
- Integrating Makefile with Other Build Tools
- Case Studies and Real-world Examples
- Simple Exercise

OBI

1. Open Bus Interface (OBI)

- Description
- Signals
- Waveform
- Operations

RTL Design

1. Introduction to RTL Design

- What is RTL (Register-Transfer Level) Design?
- Abstraction Levels in Digital Design
- Importance of RTL Design in ASIC and FPGA Development
- RTL Design Flow
- Introduction to Synthesis and FPGA/ASIC Implementation
- Learning Resources
- Exercises

2. Digital Design Fundamentals Review

- Combinational Logic
- Sequential Logic
- Timing Diagrams and Basic Timing Concepts
- Learning Resources
- Exercises

3. RTL Coding Styles and Best Practices

- Learning Resources
- Exercises

4. Clocking and Reset

- Learning Resources
- Exercises

5. Memory Design

- Learning Resources
- Exercises

6. Finite State Machines (FSMs)

- Learning Resources
- Exercises

7. Pipelining

- Learning Resources
- Exercises

8. Verification Basics

- Learning Resources
- Exercises

9. Synthesis and Implementation Basics

- Learning Resources
- Exercises

10. Advanced RTL Design Topics (Optional)

- Learning Resources
- Exercises

RTL Design Verification

1. Introduction to Verification and Verification Planning

- Learning Resources
- Exercises

2. SystemVerilog for Verification - Testbench Fundamentals

- Learning Resources
- Exercises

3. Constrained-Random Verification

- Learning Resources
- Exercises

4. Introduction to Universal Verification Methodology (UVM)

- Learning Resources
- Exercises

5. UVM Testbench Components and Advanced Features

- Learning Resources
- Exercises

6. Assertion-Based Verification (ABV) with SystemVerilog Assertions (SVA)

- Learning Resources
- Exercises

7. Debugging and Coverage Analysis in Verification

- Learning Resources
- Exercises

8. Verification Project and Advanced Verification Topics

- Learning Resources

SystemVerilog

1. SystemVerilog Foundations

- Demystifying SystemVerilog
- A Brief History of SystemVerilog
- Why Use SystemVerilog? Key Applications
- The Winning Advantages of SystemVerilog
- Core Features That Set SystemVerilog Apart

2. Basics

- I. Basic SystemVerilog Syntax Fundamentals
- II. The module Construct
- III. Procedural Blocks
- IV. The initial Block
- V. System Tasks for Simulation Control and Output

3. SystemVerilog Built-in Tasks and Functions

- Display Tasks
- Wavedump Tasks
- Time-Related Functions
- Simulation Control Tasks
- Hands-on Exercises with Solutions

4. SystemVerilog Data Types

- Built-in Data Types
- Advanced Built-in Types
- User-Defined Data Types
- Packed vs. Unpacked Arrays
- Exercises to Solidify Your Understanding

5. Arrays in SystemVerilog

- Introduction to SystemVerilog Arrays
- Packed vs. Unpacked Arrays
- Fixed-Size Arrays
- Dynamic Arrays
- Associative Arrays
- Queues
- Exercises to Practice Array Concepts

6. SystemVerilog Array Manipulation

- Introduction
- Array Indexing
- Array Manipulation Methods
- Illustrative Examples with Expected Outputs
- Important Considerations for Array Manipulation
- Practical Exercises to Enhance Your Skills
- Pro-Level Tips for Array Mastery

7. SystemVerilog Operators

- Introduction
- Arithmetic Operators
- Logical vs. Bitwise Operators
- Reduction Operators
- Comparison Operators
- Operator Precedence
- Common Pitfalls to Avoid
- Practical Exercises to Solidify Operator Skills

8. Control Flow in SystemVerilog

- Introduction
- Conditional Statements
- Case Statements
- Loop Constructs
- Exercises to Solidify Control Flow Understanding

9. Procedural Blocks in SystemVerilog

- Introduction
- Initial Blocks
- Final Blocks
- Always Blocks
- Assignment Types within Procedural Blocks
- Best Practices and Common Pitfalls with Procedural Blocks
- Exercises to Practice Procedural Blocks

10. Tasks and Functions in SystemVerilog

- Introduction
- Key Differences
- Task Implementation
- Function Implementation
- Storage Classes

- Advanced Features
- Best Practices for Tasks and Functions
- Exercises to Solidify Task and Function Concepts

11. Interprocess Communication (IPC) in SystemVerilog

- Introduction
- Mailboxes
- Semaphores
- Events
- Comparison
- Exercises
- Best Practices for Effective IPC

12. SystemVerilog Modules

- Introduction
- Module Definition
- Ports and Parameters
- Module Instantiation
- Testbench Integration
- Best Practices for SystemVerilog Module Design
- Exercises to Practice Module Design in SystemVerilog

13. SystemVerilog Interfaces

- Introduction
- Defining Interfaces
- Implementing Interfaces in Modules
- Modports
- Practical Applications and Exercises to Master Interfaces
- Best Practices and Common Pitfalls to Avoid
- Conclusion

14. SystemVerilog Randomization

- Introduction
- Random Variable Types
- Constraint Specification
- Randomization Control
- Verification Integration
- Best Practices for Effective SystemVerilog Randomization
- Exercises to Master SystemVerilog Randomization

15. SystemVerilog Classes

- Introduction

- Defining Classes
- Creating Objects
- Inheritance
- Polymorphism
- Encapsulation
- Randomization within Classes
- Exercises to Practice SystemVerilog Classes and OOP

16. SystemVerilog Packages

- Introduction
- Defining Packages
- Importing Packages
- Package Scope Resolution
- Package Export System
- Best Practices for Effective Package Usage
- Exercises to Practice SystemVerilog Packages

17. SystemVerilog Command Line Arguments

- Introduction
- Accessing Command Line Arguments
- Parameter Overriding with Command Line Arguments
- Best Practices for Command Line Arguments
- Comprehensive Function Reference Table
- Robust Error Handling for Command Line Arguments
- Best Practices Checklist for SystemVerilog Command Line Arguments
- Practical Exercises to Solidify Command Line Argument Skills

18. SystemVerilog File Operations

- Introduction
- File Handling Fundamentals
- Reading Data from Files
- Writing Data to Files
- Advanced File Output
- Comprehensive Function Reference Table
- Robust Error Handling Techniques
- Best Practices Checklist for SystemVerilog File Operations
- Practical Exercises to Solidify File Operation Skills

19. SystemVerilog Assertions (SVA)

- Introduction to Assertion-Based Verification
- Assertion Types
- Immediate Assertions

- Concurrent Assertions
- Temporal Operators
- Assertion Severity Levels
- Coverage-Driven Verification with Assertions
- Advanced Assertion Techniques
- Debugging Methodologies for SystemVerilog Assertions
- Best Practices Checklist for Effective SystemVerilog Assertion Usage
- Practical Exercises to Master SystemVerilog Assertions
- Comprehensive Function Reference Table for SystemVerilog Assertions

20. SystemVerilog Coverage

- Introduction
- Code Coverage
- Functional Coverage
- Practical Exercises to Solidify Coverage Concepts
- Best Practices for Effective Coverage-Driven Verification
- Comprehensive Reference Table

21. SystemVerilog Macros

- Introduction
- Defining and Using Macros
- Types and Applications of Macros
- Scope of Macro Definitions
- Predefined Macros
- Best Practices for Macros
- Comprehensive Directive Reference Table
- Robust Macro Usage and Error Prevention
- Best Practices Checklist for SystemVerilog Macros
- Practical Exercises to Solidify Macro Skills

22. SystemVerilog Built-in System Tasks and Functions

- Simulation Control Tasks
- Simulation Time Functions
- Timescale Tasks
- Conversion Functions
- Data Query Functions
- Array Query Functions
- Math Functions
- Bit Vector System Functions
- Severity Tasks
- Assertion Control Tasks
- Sampled Value System Functions
- Coverage Control Functions

- Probabilistic Distribution Functions
- Stochastic Analysis Tasks and Functions
- PLA Modeling Tasks
- Miscellaneous Tasks and Functions
- Display Tasks
- File I/O Tasks and Functions
- Memory Load Tasks
- Memory Dump Tasks
- Command Line Input
- VCD Tasks

Unix Terminal

1. Chapter 1

- What You Will Learn
- Key Concepts
- Essential Commands
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

2. Chapter 2

- What You Will Learn
- Key Concepts
- Essential Commands
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

3. Chapter 3

- What You Will Learn
- Key Concepts
- Essential Commands
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

4. Chapter 4

- What You Will Learn
- Key Concepts
- Essential Commands and Patterns
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

5. Chapter 5

- What You Will Learn
- Key Concepts
- Essential Commands
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

6. Chapter 6

- What You Will Learn
- Key Concepts
- Script Skeleton
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

7. Chapter 7

- What You Will Learn
- Key Concepts
- Essential Snippets
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

8. Chapter 8

- What You Will Learn
- Key Concepts
- Essential Commands and Snippets
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

9. Chapter 9

- What You Will Learn
- Key Concepts
- Essential Commands
- Exercises (Hands-On)
- Mini-Lab
- Check Your Understanding
- Common Pitfalls
- Prep for Next Chapter

10. Chapter 10

- What You Will Learn
- Capstone Ideas (Pick One)
- Suggested Workflow
- Example Skeleton (Backup)
- Exercises (Hands-On)
- Check Your Understanding
- Common Pitfalls
- Next Steps and Further Learning

VIM

1. Vi IMproved (VIM)

- Introduction
- Install
- Simple Exercise
- Study Materials

Vivado

1. Introduction to Vivado

- Features of Vivado
- Use Cases of Vivado

2. Installing Vivado

- System Requirements
- Downloading Vivado
- Installing Vivado on Windows
- Installing Vivado on Linux
- Post-Installation Steps

3. Command-Line Simulation

- Overview of Command-Line Tools
- Setting Up the Environment
- Compiling the RTL Sources
- Elaborating the Design
- Running the Simulation
- Common Options for Command-Line Tools
- Automating the Workflow with a Script

4. Simulating a Simple RTL on Vivado GUI

- Creating a New Project
- Adding RTL Sources
- Adding Simulation Sources
- Running the Simulation
- Analyzing Simulation Results

5. Synthesis in Vivado

- Setting Up the Synthesis
- Analyzing Synthesis Results
- Synthesis Settings
- Common Synthesis Options

6. Implementation in Vivado

- Setting Up the Implementation
- Analyzing Implementation Results
- Implementation Settings
- Common Implementation Options

7. Bitstream Generation in Vivado

- Generating the Bitstream
- Analyzing Bitstream Generation Results
- Programming the FPGA

8. Debugging in Vivado

- Setting Up Debug Probes
- Running the Debug
- Analyzing Debug Results

9. IP Catalog and Integration in Vivado

- Accessing the IP Catalog
- Adding an IP Core to Your Design
- Integrating the IP Core
- Generating the IP Core
- Example

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